# Coordinates and Rotation in LaRT / MoCafe

## Contents



## 1. Coordinate System in LaRT / MoCafe

- History: written on 2016-01-10; modified by Hyunmi Song on 2017-08-21; modified by KI Seon on 2017-09-01 and 2020-08-16.

### 1.1 Rotation Matrix and Peeling-off

LaRT / MoCafe can place a galaxy/cloud and an observer at arbitrary positions, depending on which side of the galaxy the observer wishes to see (face-on, edge-on, intermediate inclination, and so on). Consider two reference frames: the observer frame $O$ and the galaxy/cloud frame $G$. The target galaxy is placed at the origin with its disk lying in the $xy$-plane of $G$. Initially, $O$ and $G$ are coincident and the observer is on the $z$-axis, so by default the galaxy is seen face-on. Hence, the observer position is $(0,0,d)$ in $O$. We now rotate the observer frame so that the galaxy is viewed with a specified set of angles $(\alpha,\beta,\gamma)$. Our goal is to obtain the representation in $O$ of a vector that is given in $G$, and to determine the coordinates of the observer in $G$.

In this setting, a sequence of rotations of the observer's axes that converts the original face-on view into a view with phase angle $\alpha$, inclination angle $\beta$, and position angle $\gamma$ is, in consecutive order,

1. rotation by $\alpha$ about the $z$-axis,
2. rotation by $\beta$ about the new $y'$-axis,
3. rotation by $\gamma$ about the new $z''$-axis.

We denote the corresponding matrices by $R_z(\alpha)$, $R_y(\beta)$, and $R_z(\gamma)$, respectively, where

$$R_z(\alpha) = \begin{pmatrix} \cos\alpha & \sin\alpha & 0 \\ -\sin\alpha & \cos\alpha & 0 \\ 0 & 0 & 1 \end{pmatrix},$$

$$R_y(\beta) = \begin{pmatrix} \cos\beta & 0 & -\sin\beta \\ 0 & 1 & 0 \\ \sin\beta & 0 & \cos\beta \end{pmatrix},$$

and

$$R_z(\gamma) = \begin{pmatrix} \cos\gamma & \sin\gamma & 0 \\ -\sin\gamma & \cos\gamma & 0 \\ 0 & 0 & 1 \end{pmatrix}.$$

The composite matrix $R = R_z(\gamma)R_y(\beta)R_z(\alpha)$ rotates the frame $O$ relative to the frame $G$. Note that $R$ rotates the axes of the frame, not a vector within a fixed frame. Therefore, to re-express a given vector in the rotated frame, we multiply the vector by $R$. The matrix that gives the new coordinates of a vector in the rotated frame $O$ is therefore

$$R = R_z(\gamma)R_y(\beta)R_z(\alpha),$$

and substituting the matrices from above we obtain

$$R = \begin{pmatrix} \cos\gamma\cos\beta\cos\alpha - \sin\gamma\sin\alpha & \cos\gamma\cos\beta\sin\alpha + \sin\gamma\cos\alpha & -\cos\gamma\sin\beta \\ -\sin\gamma\cos\beta\cos\alpha - \cos\gamma\sin\alpha & -\sin\gamma\cos\beta\sin\alpha + \cos\gamma\cos\alpha & \sin\gamma\sin\beta \\ \sin\beta\cos\alpha & \sin\beta\sin\alpha & \cos\beta \end{pmatrix}.$$

The inverse matrix, which corresponds to the transformation from the observer frame to the galaxy frame, is given by

$$R^{-1} = R^T = R_z(-\alpha)R_y(-\beta)R_z(-\gamma).$$

Using this matrix, the observer's position vector in $G$ (the fixed frame), $\{\mathbf{k}\}_G$, is obtained from $R^{-1}\{\mathbf{k}\}_O$ where $\{\mathbf{k}\}_O = (0,0,d)^T$. The components of $\{\mathbf{k}\}_G$ are

$$\begin{aligned} x_{\rm obs} &= d\sin\beta\cos\alpha, \\ y_{\rm obs} &= d\sin\beta\sin\alpha, \\ z_{\rm obs} &= d\cos\beta. \end{aligned}$$

To apply the peeling-off technique, we need the direction vector from the photon position to the observer. The detector plane coincides with the $xy$-plane of the observer frame. If the observer and photon coordinates in the galaxy frame $G$ are $(x_{\rm obs}, y_{\rm obs}, z_{\rm obs})$ and $(x_{\rm p}, y_{\rm p}, z_{\rm p})$, respectively, then the photon direction toward the observer in $G$ is

$$\{\mathbf{v}\}_G = (x_{\rm obs} - x_{\rm p},\, y_{\rm obs} - y_{\rm p},\, z_{\rm obs} - z_{\rm p})^T / |\mathbf{v}|,$$

where the normalization factor is

$$|\mathbf{v}| = \left[(x_{\rm obs} - x_{\rm p})^2 + (y_{\rm obs} - y_{\rm p})^2 + (z_{\rm obs} - z_{\rm p})^2\right]^{1/2}.$$

The corresponding direction vector in the observer frame, $\{\mathbf{v}\}_O = R\{\mathbf{v}\}_G = (v_x, v_y, v_z)^T$, yields the angular coordinates $(\theta_x, \theta_y)$ of the photon in the detector $XY$-plane:

$$\theta_x = \frac{180}{\pi}\,\mathrm{atan2}(-v_x,\, v_z),$$
$$\theta_y = \frac{180}{\pi}\,\mathrm{atan2}(-v_y,\, v_z).$$

In Fortran, the array indices on the image plane with angular bin sizes $(\Delta\theta_x, \Delta\theta_y)$ are then

$$i = \mathrm{floor}\left(\frac{\theta_x}{\Delta\theta_x} + \frac{N_x}{2}\right) + 1 \quad (i = 1, \ldots, N_x),$$
$$j = \mathrm{floor}\left(\frac{\theta_y}{\Delta\theta_y} + \frac{N_y}{2}\right) + 1 \quad (j = 1, \ldots, N_y),$$

where $N_x$ and $N_y$ are the number of image bins along the $x$ and $y$ axes, respectively.

Sometimes we wish to compute the optical depth along a line of sight from a pixel on the detector toward the sky (galaxy). In that case the direction vector toward the galaxy from a pixel $(i, j)$ on the detector is given by

$$v_x = \tan\left[\left(i - \tfrac{N_x+1}{2}\right) \Delta\theta_x\, \tfrac{\pi}{180}\right] / v,$$
$$v_y = \tan\left[\left(j - \tfrac{N_y+1}{2}\right) \Delta\theta_y\, \tfrac{\pi}{180}\right] / v,$$
$$v_z = -1/v,$$

where $v = \sqrt{v_x^2 + v_y^2 + v_z^2}$ and $\Delta\theta_x, \Delta\theta_y$ are in degrees. This direction vector is computed in the detector frame and must be transformed into the galaxy frame.

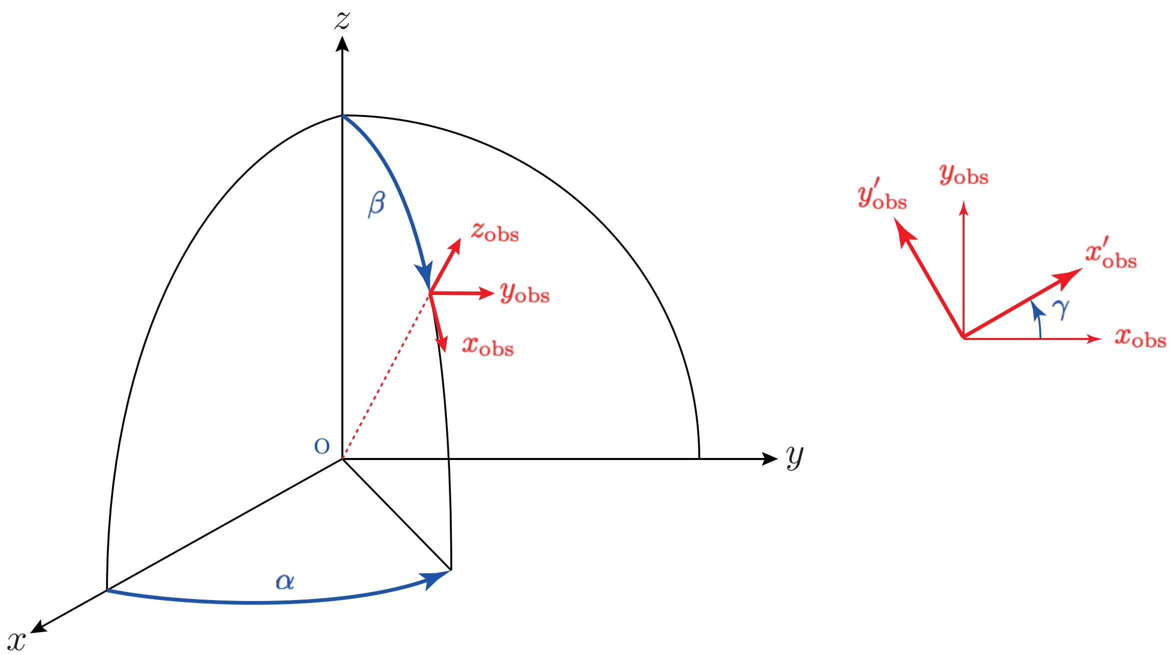



Figure 1: Coordinate-system transformation from the galaxy frame $(X, Y, Z)$ to the observer frame $(x, y, z)$.

### 1.2 Edge-on Galaxy

When modeling an edge-on galaxy, we assume that the disk lies in the $xy$-plane. The phase angle for the spiral pattern, the inclination angle of the disk, and the position angle on the sky are then given by

$$\text{phase angle} = -\alpha,$$
$$\text{inclination angle} = -\beta,$$
$$\text{position angle} = -\gamma.$$

## 2. Scattering Geometry

- History: written on 2014-12-15, updated on 2014-12-18.
- This section will be updated later (2020-08-16).

The following definitions and equations are mostly the same as those of Bianchi et al. (1996, ApJ, 465, 127). See also Chandrasekhar (1960, *Radiative Transfer*), Code & Whitney (1995, ApJ), and Whitney (2011, BASI).

### 2.1 Transformation of the Stokes Parameters

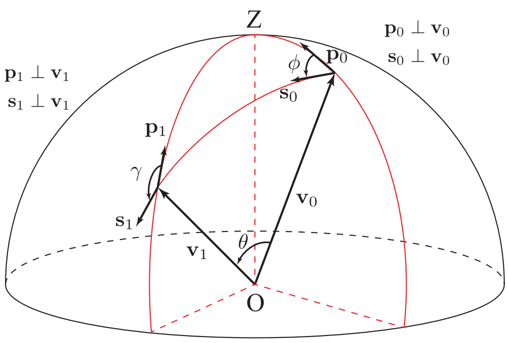



Figure 2: Scattering geometry. A photon propagating in the direction $\mathbf{v}_0$ scatters through angle $\theta$ into the direction $\mathbf{v}_1$.

We use the Stokes vector $\mathbf{S}$ to describe the polarization:

$$\mathbf{S} = \begin{pmatrix} I \\ Q \\ U \\ V \end{pmatrix},$$

where $I$ is the intensity, $Q$ is the linear polarization component aligned parallel to the $z$-axis, $U$ is the linear polarization component aligned $\pm 45°$ to the $z$-axis, and $V$ is the circular polarization. Note that the Stokes vector depends on the chosen reference frame. A scattering diagram is shown in Figure 2 (see also Chandrasekhar 1960). In the figure, $\mathbf{v}_0$ and $\mathbf{v}_1$ are unit vectors that define the propagation direction before and after scattering, respectively. The scattering plane is spanned by $\mathbf{v}_0$ and $\mathbf{v}_1$. Two angles specify $\mathbf{v}_1$ relative to $\mathbf{v}_0$: the polar angle $\theta$ between $\mathbf{v}_0$ and $\mathbf{v}_1$, which is the actual scattering angle, and the azimuthal angle $\phi$ between $\mathbf{p}_0$ and $\mathbf{s}_0$, measured clockwise about $\mathbf{v}_0$. The reference directions for the Stokes parameters of the incoming and scattered photons are $\mathbf{p}_0$ and $\mathbf{p}_1$, respectively. The unit vector $\mathbf{s}_0$ is the projection of $\mathbf{v}_1$ onto the plane normal to $\mathbf{v}_0$; $\mathbf{s}_0$ also lies in the scattering plane defined by $\mathbf{v}_0$ and $\mathbf{v}_1$.

Since the Mueller matrix elements are most naturally written for Stokes vectors referred to the scattering plane, we first rotate the Stokes vector about $\mathbf{v}_0$ from the initial reference direction $\mathbf{p}_0$ to the new reference direction $\mathbf{s}_0$. The Stokes parameters of the incoming photon referred to the scattering plane (reference direction $\mathbf{s}_0$) are then

$$\mathbf{S}_0' = \mathbf{L}(-\phi)\,\mathbf{S}_0,$$

where

$$\mathbf{L}(\varphi) = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & \cos 2\varphi & \sin 2\varphi & 0 \\ 0 & -\sin 2\varphi & \cos 2\varphi & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}.$$

The rotation matrix is defined for clockwise rotation, as in Chandrasekhar (1960, Chap. 1).

The Stokes vector after scattering, still referred to the scattering plane (reference direction $\mathbf{s}_1$), is obtained by applying the scattering Mueller matrix $\mathbf{R}(\theta)$:

$$\mathbf{S}_1' = \mathbf{R}(\theta)\,\mathbf{L}(-\phi)\,\mathbf{S}_0,$$

where

$$\mathbf{R}(\theta) = \begin{pmatrix} S_{11}(\theta) & S_{12}(\theta) & 0 & 0 \\ S_{12}(\theta) & S_{11}(\theta) & 0 & 0 \\ 0 & 0 & S_{33}(\theta) & S_{34}(\theta) \\ 0 & 0 & -S_{34}(\theta) & S_{33}(\theta) \end{pmatrix}.$$

The matrix elements can be computed with a Mie code such as `bhmie` or `MIEV0`.

The new Stokes parameters are defined relative to $\mathbf{s}_1$, the unit vector that lies in the scattering plane and in the plane perpendicular to the new propagation direction $\mathbf{v}_1$. We must therefore rotate the Stokes vector once more, from $\mathbf{s}_1$ to the final reference direction $\mathbf{p}_1$. The final Stokes vector is then

$$\mathbf{S}_1 = \mathbf{L}(\gamma)\,\mathbf{R}(\theta)\,\mathbf{L}(-\phi)\,\mathbf{S}_0,$$

and explicit multiplication of the three matrices gives

$$
\begin{aligned}
\mathbf{L}(\gamma)\mathbf{R}(\theta)\mathbf{L}(-\phi) &= \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & \cos 2\gamma & \sin 2\gamma & 0 \\ 0 & -\sin 2\gamma & \cos 2\gamma & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} S_{11} & S_{12} & 0 & 0 \\ S_{12} & S_{11} & 0 & 0 \\ 0 & 0 & S_{33} & S_{34} \\ 0 & 0 & -S_{34} & S_{33} \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & \cos 2\phi & -\sin 2\phi & 0 \\ 0 & \sin 2\phi & \cos 2\phi & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \\
&= \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & \cos 2\gamma & \sin 2\gamma & 0 \\ 0 & -\sin 2\gamma & \cos 2\gamma & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} S_{11} & S_{12}\cos 2\phi & -S_{12}\sin 2\phi & 0 \\ S_{12} & S_{11}\cos 2\phi & -S_{11}\sin 2\phi & 0 \\ 0 & S_{33}\sin 2\phi & S_{33}\cos 2\phi & S_{34} \\ 0 & -S_{34}\sin 2\phi & -S_{34}\cos 2\phi & S_{33} \end{pmatrix} \\
&= \begin{pmatrix} S_{11} & S_{12}\cos 2\phi & -S_{12}\sin 2\phi & 0 \\ S_{12}\cos 2\gamma & S_{11}\cos 2\gamma\cos 2\phi + S_{33}\sin 2\gamma\sin 2\phi & -S_{11}\cos 2\gamma\sin 2\phi + S_{33}\sin 2\gamma\cos 2\phi & S_{34}\sin 2\gamma \\ -S_{12}\sin 2\gamma & -S_{11}\sin 2\gamma\cos 2\phi + S_{33}\cos 2\gamma\sin 2\phi & S_{11}\sin 2\gamma\sin 2\phi + S_{33}\cos 2\gamma\cos 2\phi & S_{34}\cos 2\gamma \\ 0 & -S_{34}\sin 2\phi & -S_{34}\cos 2\phi & S_{33} \end{pmatrix}.
\end{aligned}
$$

## 2.2 Scattering Matrix for Resonance Scattering

The scattering matrix for resonance line scattering in the scattering plane is given by Equations (259) and (260) of Chandrasekhar (1960):

$$
\begin{pmatrix} I'_l \\ I'_r \\ U' \\ V' \end{pmatrix} = R \begin{pmatrix} I_l \\ I_r \\ U \\ V \end{pmatrix},
$$

$$
\begin{aligned}
R = \frac{3}{2}E_1 \begin{pmatrix} \cos^2\theta & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & \cos\theta & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix} &+ \frac{1}{2}E_2 \begin{pmatrix} 1 & 1 & 0 & 0 \\ 1 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix} + \frac{3}{2}E_3 \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \cos\theta \end{pmatrix} \\
= \begin{pmatrix} \frac{3}{2}E_1\cos^2\theta + \frac{1}{2}E_2 & \frac{1}{2}E_2 & 0 & 0 \\ \frac{1}{2}E_2 & \frac{3}{2}E_1 + \frac{1}{2}E_2 & 0 & 0 \\ 0 & 0 & \frac{3}{2}E_1\cos\theta & 0 \\ 0 & 0 & 0 & \frac{3}{2}E_3\cos\theta \end{pmatrix}&.
\end{aligned}
$$

Therefore,

$$
\begin{aligned}
I'_l &= \left(\tfrac{3}{2}E_1\cos^2\theta + \tfrac{1}{2}E_2\right) I_l + \tfrac{1}{2}E_2 I_r, \\
I'_r &= \tfrac{1}{2}E_2 I_l + \left(\tfrac{3}{2}E_1 + \tfrac{1}{2}E_2\right) I_r.
\end{aligned}
$$

Using the definitions $I = I_l + I_r$ and $Q = I_l - I_r$, we obtain

$$
\begin{aligned}
I' &= \left(\tfrac{3}{2}E_1\cos^2\theta + E_2\right) \tfrac{1}{2}(I+Q) + \left(\tfrac{3}{2}E_1 + E_2\right) \tfrac{1}{2}(I-Q), \\
Q' &= \left(\tfrac{3}{2}E_1\cos^2\theta\right) \tfrac{1}{2}(I+Q) - \left(\tfrac{3}{2}E_1\right) \tfrac{1}{2}(I-Q),
\end{aligned}
$$

which simplifies to

$$
\begin{aligned}
I' &= \left[\tfrac{3}{4}E_1\left(\cos^2\theta+1\right)+E_2\right]I+\tfrac{3}{4}E_1\left(\cos^2\theta-1\right)Q,\\
Q' &= \tfrac{3}{4}E_1\left(\cos^2\theta-1\right)I+\tfrac{3}{4}E_1\left(\cos^2\theta+1\right)Q.
\end{aligned}
$$

Therefore, the scattering matrix for the conventional Stokes vector defined in the scattering plane is

$$
\begin{pmatrix} I' \\ Q' \\ U' \\ V' \end{pmatrix} = \mathbf{R}(\theta)\begin{pmatrix} I \\ Q \\ U \\ V \end{pmatrix},
$$

$$
\mathbf{R}(\theta) = \tfrac{3}{4}E_1\begin{pmatrix} \cos^2\theta+1 & \cos^2\theta-1 & 0 & 0 \\ \cos^2\theta-1 & \cos^2\theta+1 & 0 & 0 \\ 0 & 0 & 2\cos\theta & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix} + E_2\begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix} + \tfrac{3}{2}E_3\begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \cos\theta \end{pmatrix}
$$

$$
\mathbf{R}(\theta) = \begin{pmatrix} \tfrac{3}{4}E_1\left(\cos^2\theta+1\right)+E_2 & \tfrac{3}{4}E_1\left(\cos^2\theta-1\right) & 0 & 0 \\ \tfrac{3}{4}E_1\left(\cos^2\theta-1\right) & \tfrac{3}{4}E_1\left(\cos^2\theta+1\right) & 0 & 0 \\ 0 & 0 & \tfrac{3}{2}E_1\cos\theta & 0 \\ 0 & 0 & 0 & \tfrac{3}{2}E_3\cos\theta \end{pmatrix}.
$$

The scattering matrix in the comoving frame is then

$$
\begin{aligned}
\mathbf{R}(\theta)\,\mathbf{L}(-\phi) &= \begin{pmatrix} \tfrac{3}{4}E_1\left(\cos^2\theta+1\right)+E_2 & \tfrac{3}{4}E_1\left(\cos^2\theta-1\right) & 0 & 0 \\ \tfrac{3}{4}E_1\left(\cos^2\theta-1\right) & \tfrac{3}{4}E_1\left(\cos^2\theta+1\right) & 0 & 0 \\ 0 & 0 & \tfrac{3}{2}E_1\cos\theta & 0 \\ 0 & 0 & 0 & \tfrac{3}{2}E_3\cos\theta \end{pmatrix}\begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & \cos 2\phi & -\sin 2\phi & 0 \\ 0 & \sin 2\phi & \cos 2\phi & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \\
&= \begin{pmatrix} \tfrac{3}{4}E_1\left(\cos^2\theta+1\right)+E_2 & \tfrac{3}{4}E_1\left(\cos^2\theta-1\right)\cos 2\phi & -\tfrac{3}{4}E_1\left(\cos^2\theta-1\right)\sin 2\phi & 0 \\ \tfrac{3}{4}E_1\left(\cos^2\theta-1\right) & \tfrac{3}{4}E_1\left(\cos^2\theta+1\right)\cos 2\phi & -\tfrac{3}{4}E_1\left(\cos^2\theta+1\right)\sin 2\phi & 0 \\ 0 & \tfrac{3}{2}E_1\cos\theta\sin 2\phi & \tfrac{3}{2}E_1\cos\theta\cos 2\phi & 0 \\ 0 & 0 & 0 & \tfrac{3}{2}E_3\cos\theta \end{pmatrix},
\end{aligned}
$$

so that

$$
\begin{aligned}
I' &= \left[\tfrac{3}{4}E_1\left(\cos^2\theta+1\right)+E_2\right]I+\left[\tfrac{3}{4}E_1\left(\cos^2\theta-1\right)\cos 2\phi\right]Q-\left[\tfrac{3}{4}E_1\left(\cos^2\theta-1\right)\sin 2\phi\right]U,\\
Q' &= \left[\tfrac{3}{4}E_1\left(\cos^2\theta-1\right)\right]I+\left[\tfrac{3}{4}E_1\left(\cos^2\theta+1\right)\cos 2\phi\right]Q-\left[\tfrac{3}{4}E_1\left(\cos^2\theta+1\right)\sin 2\phi\right]U,\\
U' &= \left[\tfrac{3}{2}E_1\cos\theta\sin 2\phi\right]Q+\left[\tfrac{3}{2}E_1\cos\theta\cos 2\phi\right]U,\\
V' &= \left[\tfrac{3}{2}E_3\cos\theta\right]V.
\end{aligned}
$$

## 2.3 Polarization Handedness and Angle

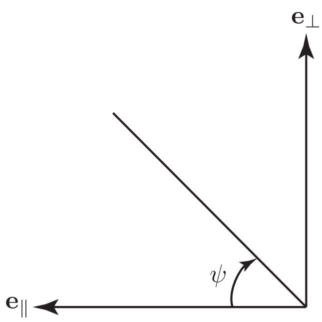
The linear polarization map is obtained by computing, for each pixel, the degree of linear polarization and the polarization angle:

$$P = \sqrt{Q^2 + U^2}/I,$$
$$\tan 2\psi = U/Q,$$

or equivalently

$$Q/I = P\cos 2\psi,$$
$$U/I = P\sin 2\psi.$$

We adopt the traditional convention in which an elliptically polarized wave is right-handed if the vibration ellipse rotates clockwise as viewed by an observer looking toward the source of light (Figure 3). The opposite convention seems to be favored by van de Hulst (1957). The sign of $V$ specifies the handedness of the vibration ellipse: positive denotes right-handed and negative denotes left-handed. The polarization angle $\psi$ is defined as the clockwise angle between $\mathbf{e}_{\|}$ and $\mathbf{e}_{\perp}$.




*Linearly Polarized*

| $0^\circ$ $\leftrightarrow$ | $90^\circ$ $\updownarrow$ | $+45^\circ$ $\nearrow$ | $-45^\circ$ $\searrow$ | $\gamma$ |
|---|---|---|---|---|
| $\begin{pmatrix}1\\1\\0\\0\end{pmatrix}$ | $\begin{pmatrix}1\\-1\\0\\0\end{pmatrix}$ | $\begin{pmatrix}1\\0\\1\\0\end{pmatrix}$ | $\begin{pmatrix}1\\0\\-1\\0\end{pmatrix}$ | $\begin{pmatrix}1\\\cos 2\gamma\\\sin 2\gamma\\0\end{pmatrix}$ |

*Circularly Polarized*

| Right $\circlearrowright$ | Left $\circlearrowleft$ |
|---|---|
| $\begin{pmatrix}1\\0\\0\\1\end{pmatrix}$ | $\begin{pmatrix}1\\0\\0\\-1\end{pmatrix}$ |

Figure 3: (Left) Definition of the polarization angle and handedness. (Right) Stokes parameters for polarized light, reproduced from Table 2.2 of Bohren & Huffman (1983). The unit vectors $\mathbf{e}_{\|}$ and $\mathbf{e}_{\perp}$ correspond to the $\mathbf{y}$ and $\mathbf{x}$ axes, respectively.

## 2.4 Phase Function

### 2.4.1 Scattering Angles

The phase function follows from $\mathbf{S}_1' = \mathbf{R}(\theta)\,\mathbf{L}(-\phi)\,\mathbf{S}_0$:

$$\begin{aligned} P(\theta,\phi) &= \frac{I'(\theta,\phi)}{I} = S_{11}(\theta) + S_{12}(\theta)\,\frac{Q'}{I} \\ &= S_{11}(\theta) + S_{12}(\theta)\left(\frac{Q}{I}\cos 2\phi - \frac{U}{I}\sin 2\phi\right). \end{aligned}$$

The angles $\theta$ and $\phi$ are distributed according to $P(\theta,\phi)$, which is normalized so that

$$\begin{aligned}1 &= \int_0^{2\pi} d\phi \int_0^{\pi} d\theta \sin\theta \, P(\theta,\phi) \\ &= \int_0^{\pi} d\theta \sin\theta \, P(\theta),\end{aligned}$$

where

$$P(\theta) = \int_0^{2\pi} P(\theta,\phi)\, d\phi = S_{11}(\theta).$$

Because $P(\theta)$ depends only on $\theta$, the scattering polar angle $\bar{\theta}$ can be drawn from a single random number $\xi_1$ independently of both $\phi$ and the polarization of the incident radiation, by inverting

$$\int_0^{\bar{\theta}} P(\theta) \sin\theta \, d\theta = \xi_1.$$

Substituting $\bar{\theta}$ into the equation for the joint distribution, $\bar{\phi}$ is then obtained from

$$\int_0^{\bar{\phi}} P(\bar{\theta},\phi)\, d\phi \Bigg/ \int_0^{2\pi} P(\bar{\theta},\phi)\, d\phi = \xi_2,$$

or, explicitly,

$$\frac{1}{2\pi}\left\{\bar{\phi} + \frac{S_{12}(\bar{\theta})}{2S_{11}(\bar{\theta})}\left[\frac{Q}{I}\sin 2\bar{\phi} - \frac{U}{I}\left(1-\cos 2\bar{\phi}\right)\right]\right\} = \xi_2.$$

In terms of the polarization angle $\chi$ defined by $Q/I = \cos 2\beta \cos 2\chi$ and $U/I = \cos 2\beta \sin 2\chi$, the phase function and the equation for $\bar{\phi}$ become

$$\begin{aligned}P(\theta,\phi) &= S_{11}(\theta) + S_{12}(\theta)\cos 2\beta \cos 2(\phi+\chi), \\ P(\theta) &= S_{11}(\theta), \\ P(\phi) &= \frac{1}{2\pi}\left[1 + \frac{S_{12}(\bar{\theta})}{S_{11}(\bar{\theta})}\cos 2\beta \cos 2(\phi+\chi)\right].\end{aligned}$$

For a rejection method, the maximum of $P(\phi)$ is

$$P_{\max} = \frac{1}{2\pi}\left[1 + \left|\frac{S_{12}}{S_{11}} P_L\right|\right],$$

where $P_L = \cos 2\beta$ measures the linear polarization fraction. Note that $\cos 2\beta = 1$ for linearly polarized light.

The equation for $\bar{\phi}$ can be rewritten as

$$f(\bar{\phi}) \equiv \frac{1}{2\pi}\left\{\bar{\phi} + \frac{S_{12}(\bar{\theta})}{2S_{11}(\bar{\theta})}\cos 2\beta\left[\sin 2(\bar{\phi}+\chi) - \sin 2\chi\right]\right\} = \xi_2.$$

How do we solve this equation for given $\bar{\theta}$ and $\xi_2$? Note that $|S_{12}| \le |S_{11}|$ and $0 \le P_L \le 1$. The ratio $S_{12}/S_{11}$ can reach 100% at $\bar{\theta} = \pi/2$ for Rayleigh scattering at long wavelengths ($\lambda \gtrsim 10\,\mu$m); however, in most practical cases $|S_{12}/S_{11}| < 1$. The derivative of $f$ is

$$f'(\phi) = \frac{1}{2\pi}\left\{1 + \frac{S_{12}}{S_{11}} P_L \cos 2(\phi+\chi)\right\} \ge 0.$$

A value of $\phi$ with $f'(\phi) = 0$ exists only when $|S_{12}/S_{11}|\, P_L = 1$, in which case the Newton–Raphson method is inapplicable. In most cases $|S_{12}/S_{11}| < 1$ and $P_L < 1$, so the Newton–Raphson method can be used; we start the iteration from $\bar{\phi} = 2\pi\xi_2$. If $f'(\phi) = 0$ is encountered, Brent's method should be used instead.

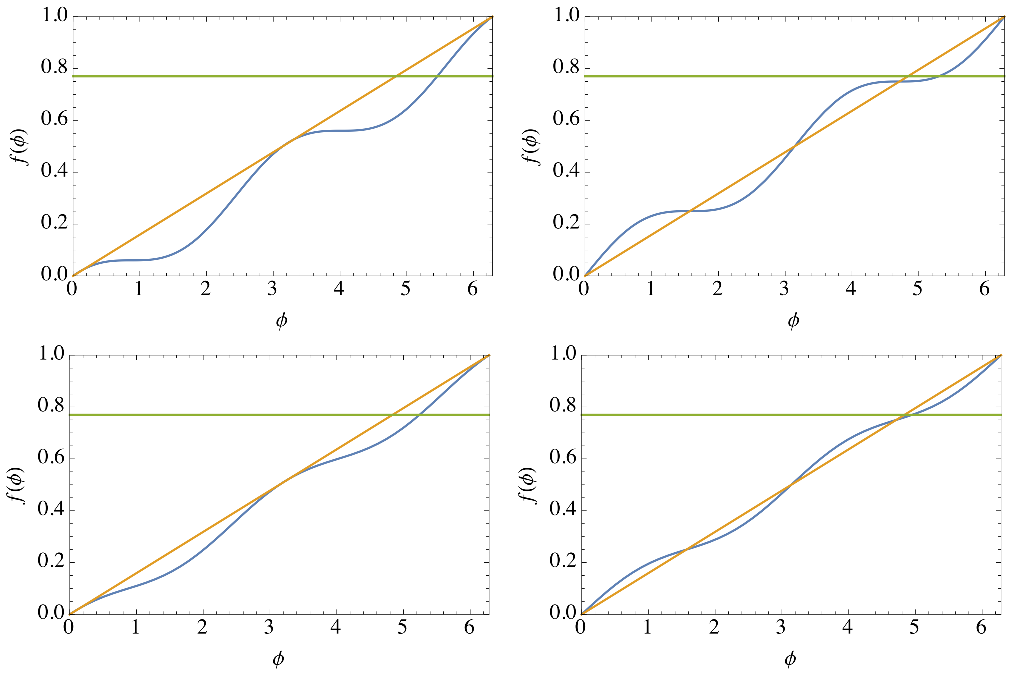



Figure 4: The function $f(\bar{\phi})$ used to compute the azimuthal scattering angle $\bar{\phi}$. Blue and orange curves denote $y = f(\bar{\phi})$ and $y = \bar{\phi}/2\pi$, respectively; green curves denote $y = \xi_2 = 0.77$. Top panels show $|S_{12}/S_{11}|P_L = 1$, bottom panels show $|S_{12}/S_{11}|P_L = 0.5$. The polarization angle was set to $\chi = 0.7$ in the left panels and $\chi = 0$ in the right panels. These plots can be reproduced with the Mathematica notebook `phase_phi.nb`.

### 2.4.2 New Direction Cosines

Given $\bar{\theta}$ and $\bar{\phi}$, we obtain the direction cosines of $\mathbf{v}_1 = (v_x^1, v_y^1, v_z^1)$ as follows. Introduce a unit vector $\mathbf{w}_0$ that lies in the $x$–$y$ plane and forms an orthonormal triad with $\mathbf{v}_0$ and $\mathbf{p}_0$ (see the left panel of Figure 5); that is, $\mathbf{w}_0 \cdot \mathbf{v}_0 = 0$ and $\mathbf{w}_0 \cdot \hat{\mathbf{z}} = 0$:

$$\begin{aligned}\mathbf{w}_0 &= \frac{1}{\rho_0}\mathbf{v}_0 \times \hat{\mathbf{z}} \\ &= \left(v_y^0\,\hat{\mathbf{x}} - v_x^0\,\hat{\mathbf{y}}\right)/\rho_0,\end{aligned}$$

where $\rho_0 = |\mathbf{v}_0 \times \hat{\mathbf{z}}| = \sqrt{(v_x^0)^2 + (v_y^0)^2} = \sqrt{1-(v_z^0)^2}$. The unit vector $\mathbf{p}_0$ is then

$$\begin{aligned}\mathbf{p}_0 = \mathbf{w}_0 \times \mathbf{v}_0 &= \frac{1}{\rho_0}\left(\mathbf{v}_0 \times \hat{\mathbf{z}}\right) \times \mathbf{v}_0 \\ &= \frac{1}{\rho_0}\left\{(\mathbf{v}_0\cdot\mathbf{v}_0)\,\hat{\mathbf{z}} - (\hat{\mathbf{z}}\cdot\mathbf{v}_0)\,\mathbf{v}_0\right\} \\ &= \frac{1}{\rho_0}\left\{\hat{\mathbf{z}} - (\hat{\mathbf{z}}\cdot\mathbf{v}_0)\,\mathbf{v}_0\right\} \\ &= \left(-\frac{v_x^0 v_z^0}{\rho_0}\right)\hat{\mathbf{x}} + \left(-\frac{v_y^0 v_z^0}{\rho_0}\right)\hat{\mathbf{y}} + \rho_0\,\hat{\mathbf{z}}.\end{aligned}$$

The vector $\mathbf{v}_1$ can be expressed in terms of the orthonormal triad ($\mathbf{p}_0$, $\mathbf{w}_0$, $\mathbf{v}_0$):

$$\mathbf{v}_1 = \sin\bar{\theta}\left(\cos\bar{\phi}\,\mathbf{p}_0 + \sin\bar{\phi}\,\mathbf{w}_0\right) + \cos\bar{\theta}\,\mathbf{v}_0.$$

Substituting the expressions for $\mathbf{p}_0$, $\mathbf{w}_0$, and $\mathbf{v}_0$ yields the components of $\mathbf{v}_1$ in the stationary coordinate system:

$$
\begin{aligned}
v_x^1 &= \frac{\sin\bar{\theta}}{\rho_0}\left(-v_x^0 v_z^0 \cos\bar{\phi} + v_y^0 \sin\bar{\phi}\right) + \cos\bar{\theta}\, v_x^0, \\
v_y^1 &= \frac{\sin\bar{\theta}}{\rho_0}\left(-v_y^0 v_z^0 \cos\bar{\phi} - v_x^0 \sin\bar{\phi}\right) + \cos\bar{\theta}\, v_y^0, \\
v_z^1 &= \rho_0 \sin\bar{\theta}\cos\bar{\phi} + \cos\bar{\theta}\, v_z^0.
\end{aligned}
$$

These are exactly Equations (A37) of Bianchi et al. (1996). Note that $\bar{\phi}$ is related to the angle $\varphi'$ defined in Pozdnyakov et al. (1983, *Soviet Scientific Reviews*, p. 323) by $\varphi' = \bar{\phi} - \pi/2$.

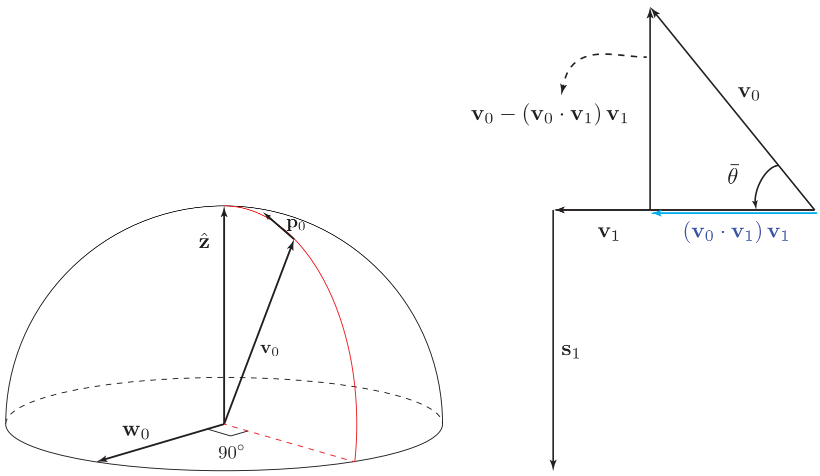



Figure 5: Geometry of (a) $\mathbf{w}_0$ and $\mathbf{p}_0$, and (b) the vectors $\mathbf{v}_0$, $\mathbf{v}_1$, and $\mathbf{s}_1$.

### 2.4.3 Transformation Angle for the Stokes Parameters

The expression for $\mathbf{p}_1$ is analogous to that for $\mathbf{p}_0$ with $\mathbf{v}_1$ in place of $\mathbf{v}_0$:

$$
\begin{aligned}
\mathbf{p}_1 &= \frac{1}{\rho_1}\left(\mathbf{v}_1 \times \hat{\mathbf{z}}\right) \times \mathbf{v}_1 \\
&= \frac{1}{\sqrt{1-(v_z^1)^2}}\left\{\hat{\mathbf{z}} - (\hat{\mathbf{z}}\cdot\mathbf{v}_1)\,\mathbf{v}_1\right\},
\end{aligned}
$$

where $\rho_1 = \sqrt{1-(v_z^1)^2}$. Because $\mathbf{s}_1$ lies in the scattering plane, it is parallel to the projection of $\mathbf{v}_0$ onto the plane normal to $\mathbf{v}_1$ but points in the opposite direction (see the right panel of Figure 5):

$$
\begin{aligned}
\mathbf{s}_1 &\parallel (\mathbf{v}_0\cdot\mathbf{v}_1)\,\mathbf{v}_1 - \mathbf{v}_0 \\
&\parallel (\mathbf{v}_0 \times \mathbf{v}_1) \times \mathbf{v}_1.
\end{aligned}
$$

Since the magnitude of $(\mathbf{v}_0\cdot\mathbf{v}_1)\mathbf{v}_1 - \mathbf{v}_0$ is $\sin\bar{\theta} = \sqrt{1-(\mathbf{v}_0\cdot\mathbf{v}_1)^2}$, the unit vector $\mathbf{s}_1$ is given by

$$
\begin{aligned}
\mathbf{s}_1 &= \frac{1}{\sqrt{1-(\mathbf{v}_0\cdot\mathbf{v}_1)^2}}\left\{(\mathbf{v}_0\cdot\mathbf{v}_1)\,\mathbf{v}_1 - \mathbf{v}_0\right\} \\
&= \frac{1}{\sqrt{1-(\mathbf{v}_0\cdot\mathbf{v}_1)^2}}(\mathbf{v}_0 \times \mathbf{v}_1) \times \mathbf{v}_1.
\end{aligned}
$$

The angle $\gamma$ between $\mathbf{p}_1$ and $\mathbf{s}_1$ is then

$$\begin{aligned}
\cos\gamma &= \mathbf{p}_1 \cdot \mathbf{s}_1 \\
&= \frac{1}{\sqrt{1-(v_z^1)^2}\sqrt{1-(\mathbf{v}_0\cdot\mathbf{v}_1)^2}} \left\{ (\mathbf{v}_0\cdot\mathbf{v}_1)(\hat{\mathbf{z}}\cdot\mathbf{v}_1) - \hat{\mathbf{z}}\cdot\mathbf{v}_0 \right\} \\
&= \frac{v_z^1 (\mathbf{v}_0\cdot\mathbf{v}_1) - v_z^0}{\sqrt{1-(v_z^1)^2}\sqrt{1-(\mathbf{v}_0\cdot\mathbf{v}_1)^2}}.
\end{aligned}$$

Since both $\mathbf{s}_1$ and $\mathbf{p}_1$ are perpendicular to $\mathbf{v}_1$, we obtain

$$\begin{aligned}
(\sin\gamma)\,\mathbf{v}_1 &= \mathbf{p}_1 \times \mathbf{s}_1 \\
&= \frac{1}{\sqrt{1-(v_z^1)^2}\sqrt{1-(\mathbf{v}_0\cdot\mathbf{v}_1)^2}} \left[ \left(\mathbf{v}_1 \times \hat{\mathbf{z}}\right) \times \mathbf{v}_1 \right] \times \left[ (\mathbf{v}_0 \times \mathbf{v}_1) \times \mathbf{v}_1 \right].
\end{aligned}$$

Using the identity $(\mathbf{A}\times\mathbf{B})\times(\mathbf{C}\times\mathbf{D}) = [\mathbf{A}\cdot(\mathbf{B}\times\mathbf{D})]\,\mathbf{C} - [\mathbf{A}\cdot(\mathbf{B}\times\mathbf{C})]\,\mathbf{D}$, we find

$$\begin{aligned}
\left[ \left(\mathbf{v}_1 \times \hat{\mathbf{z}}\right) \times \mathbf{v}_1 \right] \times \left[ (\mathbf{v}_0 \times \mathbf{v}_1) \times \mathbf{v}_1 \right] &= -\left\{ \left(\mathbf{v}_1 \times \hat{\mathbf{z}}\right) \cdot \left[ \mathbf{v}_1 \times (\mathbf{v}_0 \times \mathbf{v}_1) \right] \right\} \mathbf{v}_1 \\
&= -\left\{ \left(\mathbf{v}_1 \times \hat{\mathbf{z}}\right) \cdot \left[ \mathbf{v}_0 - (\mathbf{v}_0\cdot\mathbf{v}_1)\,\mathbf{v}_1 \right] \right\} \mathbf{v}_1 \\
&= -\left[ \left(\mathbf{v}_1 \times \hat{\mathbf{z}}\right) \cdot \mathbf{v}_0 \right] \mathbf{v}_1 \\
&= -\left[ \hat{\mathbf{z}} \cdot (\mathbf{v}_0 \times \mathbf{v}_1) \right] \mathbf{v}_1,
\end{aligned}$$

where we used the identity $\mathbf{A}\cdot(\mathbf{B}\times\mathbf{C}) = \mathbf{B}\cdot(\mathbf{C}\times\mathbf{A}) = \mathbf{C}\cdot(\mathbf{A}\times\mathbf{B})$. We finally obtain

$$\begin{aligned}
\sin\gamma &= \frac{-\hat{\mathbf{z}}\cdot(\mathbf{v}_0 \times \mathbf{v}_1)}{\sqrt{1-(v_z^1)^2}\sqrt{1-(\mathbf{v}_0\cdot\mathbf{v}_1)^2}} \\
&= \frac{v_y^0 v_x^1 - v_x^0 v_y^1}{\sqrt{1-(v_z^1)^2}\sqrt{1-(\mathbf{v}_0\cdot\mathbf{v}_1)^2}}.
\end{aligned}$$

## 2.5 Stokes Parameters in Monte Carlo Simulation

In a Monte Carlo simulation the scattered photon is not distributed over the entire solid angle but is instead deflected into a single direction and attenuated by a factor equal to the albedo. The Stokes vector after scattering is therefore

$$\tilde{\mathbf{S}}_1 = \frac{aI_0}{I_1}\mathbf{S}_1 = \frac{aI_0}{I_1}\mathbf{L}(\gamma)\,\mathbf{R}(\theta)\,\mathbf{L}(-\phi)\,\mathbf{S}_0,$$

or

$$\begin{pmatrix} \tilde{I}_1 \\ \tilde{Q}_1 \\ \tilde{U}_1 \\ \tilde{V}_1 \end{pmatrix} = \frac{aI_0}{I_1} \begin{pmatrix} I_1 \\ Q_1 \\ U_1 \\ V_1 \end{pmatrix}.$$